

# The Spectral Geometry of Misalignment

Aryan Gupta  
aryan.cs.app@gmail.com

June 2026

## Abstract

Given a fine-tuned model and a matched benign control, we ask whether the *spectrum* of the weight update reveals a concentrated component associated with a misaligned objective, without fitting the statistic on behavioral labels. We model the weight increment  $\Delta W = W_{\text{ft}} - W_{\text{base}}$  of a single layer as an iid Gaussian rectangular matrix plus a deterministic fixed-rank signal, and apply the additive rectangular singular-value transition of Benaych-Georges and Nadakuditi [8]. The classical Baik–Ben Arous–Péché (BBP) spiked-covariance results [5, 6, 7] provide historical context but concern a different, multiplicative Gaussian covariance model. In the additive model, a direction of strength  $\theta$  detaches from the Marchenko–Pastur bulk [1] if and only if  $\theta > \sqrt{\gamma}$ , where  $\gamma$  is the layer aspect ratio. At fixed signal energy  $\tau$  and near-uniform allocation, a rank- $r$  update is detectable when  $r < r_* = \tau/\sqrt{\gamma}$ . Above threshold, the leading sample eigenvector has computable overlap with the planted direction and provides a label-free direction estimate. A known-scale Tracy–Widom null [9, 4] controls false positives under the Gaussian asymptotic model; a fitted scale requires separate calibration. These conclusions assume that a misaligned update concentrates its energy in fewer directions than an energy-matched benign update. Whether real fine-tunes satisfy that assumption is the principal empirical question. We give a label-free estimator and characterize its main confounds and the limits of applying it to transformer activations.

## 1 Introduction

Supervised white-box probes can read deception from residual-stream activations with high accuracy when labeled examples from the relevant distribution are available [31]. We consider the setting in which those labels and that distribution are unavailable. In one four-team audit, the three teams with model-and-data access succeeded and the API-only team failed; differing access conditions and unusually searchable synthetic documents prevent attributing the result to any single resource [32].

We therefore analyze the weight increment  $\Delta W = W_{\text{ft}} - W_{\text{base}}$  directly. Its spectrum supplies a same-recipe screening statistic that can be calibrated against a matched benign control before behavioral validation.

Heavy-tailed self-regularization reads a label-free, data-free fingerprint of model quality directly from weight spectra, including a “bulk plus spikes” regime [12]. Spectral outlier analysis detects data poisoning from the singular structure of learned representations [30]. Random-matrix methods have also been applied to transformer internals for hallucination detection [29], while related work connects weight spectra, activation covariance, and fine-tuning in language models [14]. We ask whether update concentration separates matched benign and misaligned weight updates.

**Results and scope.** The phase transition, eigenvalue location, and overlap formula are established results [8]. This note specializes them to checkpoint increments, derives the fixed-energy rank corollary, and formulates a matched-control protocol; empirical validity remains a separate question. The main results are:

1. A random-matrix model of fine-tuning as a deterministic fixed-rank perturbation of an iid Gaussian layer increment, with the diffuse part playing the role of the Marchenko–Pastur bulk [1] and the additive rectangular transition governed by [8] (Section 4).
2. A detectability theorem: a signal direction of strength  $\theta$  is spectrally visible exactly when  $\theta > \sqrt{\gamma}$ , with the eigenvalue location given in closed form (Proposition 5.1).
3. A *rank-at-fixed-energy* result: for equal-strength fixed-rank signals, detectability is equivalent to  $r < r_\star = \tau/\sqrt{\gamma}$ ; for arbitrary allocations, low rank gives a one-sided guarantee and high rank does not imply invisibility (Proposition 5.2).
4. A label-free identification result: above threshold the leading sample eigenvector estimates the planted direction with computable overlap (Proposition 5.4), and a known-scale Gaussian Tracy–Widom test with ideal-null false-positive control (Proposition 5.5).
5. A subspace observable, the principal-angle distance between the leading eigenspaces of matched models, intended to capture signal carried by eigenvector rotation rather than eigenvalue magnitude (Definition 5.6, Remark 5.7), with surrogate calibration as a complement to scalar spectral summaries.
6. A characterization of the main failure modes (Section 8), including assumption violations for weights and activations, the limitations of activation covariance, the relation between weight-update rank and activation effective rank, and a matched-concentration test of the hypothesis.

**Theoretical and empirical scope.** Section 5 establishes an ideal-model implication: if a misaligned fine-tune deposits a rank- $r_m$  signal and an equal-energy benign fine-tune spreads its perturbation over a fixed rank  $r_b \gg r_m$ , then the spectral test separates them at an explicit critical rank. The concentration hypothesis is motivated by evidence that emergent misalignment is mediated by a single, convergent linear direction [19], a single rank-one adapter suffices to induce it [18], and emergent-misalignment weight deltas occupy a shared low-dimensional subspace across tasks [22]. Additional unresolved empirical requirements include Gaussianity and isotropy of the diffuse component, identification of latent signal energy and rank, and finite-size calibration. Section 7 describes observable proxies rather than treating these quantities as known.

**Spectral scope.** The project name reflects an analogy with Fourier inspection of vision models. Weight matrices lack a canonical periodic axis, so the analysis here uses singular values and random-matrix theory. Fourier methods require additional group structure and fall outside the present static-detection setting.

## 2 Setup and notation

Fix a layer with weight matrix  $W \in \mathbb{R}^{p \times q}$ , and write  $W_{\text{base}}$  for its value in a reference model and  $W_{\text{ft}}$  for its value after fine-tuning. The object of study is the increment

$$\Delta W = W_{\text{ft}} - W_{\text{base}} \in \mathbb{R}^{p \times q}.$$

Without loss of generality take  $q \leq p$  (otherwise transpose), and set the aspect ratio

$$\gamma = q/p \in (0, 1].$$

We treat the  $p$  rows of  $\Delta W$  as  $p$  observations in  $\mathbb{R}^q$  and form the symmetric positive semidefinite matrix

$$C = \frac{1}{p} \Delta W^\top \Delta W \in \mathbb{R}^{q \times q}.$$

The nonzero eigenvalues of  $C$  are the squared singular values of  $\Delta W$  divided by  $p$ . We write  $\lambda_1(C) \geq \dots \geq \lambda_q(C) \geq 0$  for its eigenvalues and  $\hat{v}_1, \dots, \hat{v}_q$  for the corresponding unit eigenvectors.

**Orientation.** The covariance  $C = \Delta W^\top \Delta W / p$  is input-side: its eigenvectors are the right singular vectors of  $\Delta W$ . For residual-stream interventions in the empirical paper we use the output-side covariance  $\tilde{C} = \Delta W \Delta W^\top / q$ , whose eigenvectors are the left singular vectors. Transposition preserves singular values and algebraic rank, but the displayed covariance normalizations differ: the nonzero eigenvalues of  $\tilde{C}$  are  $p/q$  times those of  $C$ . Its null edge and fluctuation scale must therefore be rescaled accordingly. The dimensionless additive detectability condition is equivalent after transposition. We state all scalar formulas in the  $q \leq p$ , input-side normalization, while behavioral interventions use output-side singular vectors.

**Energy.** The *energy* of the update is its squared Frobenius norm,  $\|\Delta W\|_F^2 = \text{Tr}(\Delta W^\top \Delta W) = p \text{Tr}(C)$ . This observed total-update energy is distinct from the latent signal energy  $\|S\|_F^2$  in Definition 4.1. Matching the former controls overall movement but does not identify or equalize the latter.

**Matched models.** The comparison uses three models sharing a base:  $W_{\text{base}}$ ; a benign control  $W_{\text{b}}$  obtained by fine-tuning on an alignment-neutral objective; and a misaligned model  $W_{\text{m}}$  obtained by an otherwise identical recipe that installs the misaligned objective. Write  $\Delta W_{\text{b}} = W_{\text{b}} - W_{\text{base}}$  and  $\Delta W_{\text{m}} = W_{\text{m}} - W_{\text{base}}$ , with associated  $C_{\text{b}}, C_{\text{m}}$ . The control is energy-matched,  $\|\Delta W_{\text{b}}\|_F = \|\Delta W_{\text{m}}\|_F$ , which controls total update magnitude but does not isolate latent signal energy, noise scale, or every training difference. Because any fine-tune can move the spectrum, comparison with the base alone cannot attribute that movement to misalignment.

### 3 Random-matrix preliminaries

The following standard results use the normalization of Section 2. Throughout, the asymptotic regime is  $p, q \rightarrow \infty$  with  $q/p \rightarrow \gamma \in (0, 1]$ .

#### 3.1 The Gaussian bulk

**Theorem 3.1** (Marchenko–Pastur law and Gaussian edge [1, 2]). *Let  $X \in \mathbb{R}^{p \times q}$  have iid  $\mathcal{N}(0, \sigma^2)$  entries and let  $S = \frac{1}{p} X^\top X$ . As  $p, q \rightarrow \infty$  with  $q/p \rightarrow \gamma \in (0, 1]$ , the empirical spectral distribution of  $S$  converges almost surely to the Marchenko–Pastur law supported on  $[\lambda_-, \lambda_+]$ , where*

$$\lambda_{\pm} = \sigma^2(1 \pm \sqrt{\gamma})^2.$$

*The largest eigenvalue also converges almost surely to  $\lambda_+ = \sigma^2(1 + \sqrt{\gamma})^2$ .*

The upper edge is the asymptotic reference level for the ideal Gaussian null. For finite matrices, an edge exceedance is descriptive until it is compared with the null fluctuation scale of Theorem 3.5.

### 3.2 The additive rectangular spike model

**Theorem 3.2** (Rectangular additive spike [8]). *Let  $N \in \mathbb{R}^{p \times q}$  have iid  $\mathcal{N}(0, \sigma^2)$  entries, with  $q/p \rightarrow \gamma \in (0, 1]$ . Let  $a \in \mathbb{R}^p$  and  $b \in \mathbb{R}^q$  be deterministic unit vectors, set  $S = \beta ab^\top$ , and define  $\theta = \beta^2/(p\sigma^2)$ . For  $C = p^{-1}(N + S)^\top(N + S)$ ,*

$$\lambda_1(C) \xrightarrow{a.s.} \begin{cases} \sigma^2(1 + \theta) \left(1 + \frac{\gamma}{\theta}\right), & \theta > \sqrt{\gamma}, \\ \sigma^2(1 + \sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases}$$

*If  $\hat{b}_1$  is the leading right singular vector of  $N + S$ , then*

$$|\langle \hat{b}_1, b \rangle|^2 \xrightarrow{a.s.} \begin{cases} \frac{1 - \gamma/\theta^2}{1 + \gamma/\theta}, & \theta > \sqrt{\gamma}, \\ 0, & \theta \leq \sqrt{\gamma}. \end{cases}$$

The Gaussian specialization permits deterministic signal directions because the noise is bi-orthogonally invariant. The general theory also treats fixed-rank perturbations under explicit isotropy and edge assumptions. Growing-rank signals and correlated noise require additional theory.

### 3.3 Historical covariance analogue

The classical Baik–Ben Arous–Péché literature [5, 6, 7] studies a different model: a low-rank perturbation of a population covariance [4]. The following Gaussian simple-spike statements explain the historical terminology and provide the multiplicative covariance analogue.

**Theorem 3.3** (Classical spiked-covariance eigenvalue limit [6]). *Let the rows of  $Y \in \mathbb{R}^{p \times q}$  be iid  $\mathcal{N}(0, \Sigma)$ , where  $\Sigma = \sigma^2(I_q + \theta vv^\top)$ ,  $v$  is a deterministic unit vector, and the population spike is simple. For  $S_{\text{cov}} = p^{-1}Y^\top Y$  and  $q/p \rightarrow \gamma \in (0, 1]$ ,*

$$\lambda_1(S_{\text{cov}}) \xrightarrow{a.s.} \begin{cases} \sigma^2(1 + \theta) \left(1 + \frac{\gamma}{\theta}\right), & \theta > \sqrt{\gamma}, \\ \sigma^2(1 + \sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases}$$

**Theorem 3.4** (Classical spiked-covariance eigenvector overlap [7]). *Under the Gaussian simple-spike conditions of Theorem 3.3, the leading sample eigenvector  $\hat{v}_1$  satisfies*

$$|\langle \hat{v}_1, v \rangle|^2 \xrightarrow{a.s.} \begin{cases} \frac{1 - \gamma/\theta^2}{1 + \gamma/\theta}, & \theta > \sqrt{\gamma}, \\ 0, & \theta \leq \sqrt{\gamma}. \end{cases}$$

The additive and multiplicative models share these white-noise formulas while retaining different probability laws.

### 3.4 The known-scale null and a qualitative general bulk

**Theorem 3.5** (Known-scale Gaussian Tracy–Widom edge [9, 4]). *For  $S = p^{-1}N^\top N$  with iid real  $N_{ij} \sim \mathcal{N}(0, \sigma^2)$ , known  $\sigma^2$ , and  $q/p \rightarrow \gamma \in (0, 1]$ ,*

$$\frac{\lambda_1(S) - \mu_{p,q}}{s_{p,q}} \Rightarrow \text{TW}_1,$$

where  $\text{TW}_1$  is the order-one Tracy–Widom law and

$$\mu_{p,q} = \frac{\sigma^2}{p}(\sqrt{p-1} + \sqrt{q})^2, \quad s_{p,q} = \frac{\sigma^2}{p}(\sqrt{p-1} + \sqrt{q}) \left( \frac{1}{\sqrt{p-1}} + \frac{1}{\sqrt{q}} \right)^{1/3}.$$

The law was introduced for orthogonal ensembles by Tracy and Widom; Johnstone established this real Wishart/PCA scaling.

**Remark 3.6** (Qualitative rectangular D-transform fallback). For a non-white rectangular noise matrix with a compact limiting singular-value law, no stray noise singular values, and the invariance or isotropic-direction hypotheses of [8], a deterministic fixed-rank perturbation has a qualitative outlier transition governed by the rectangular  $D$ -transform of the noise singular-value law. The outlier location and threshold must be computed from that law rather than from the Marchenko–Pastur closed forms. Applying this result to checkpoint increments requires empirical checks of its invariance and edge assumptions.

## 4 A spiked model of fine-tuning

We model the weight increment as a diffuse component plus a deterministic fixed-rank component, with assumptions that give the diffuse term a Marchenko–Pastur bulk [1].

**Definition 4.1** (Diffuse-plus-signal decomposition). Write the increment as

$$\Delta W = N + S, \quad S = \sum_{i=1}^r \beta_i a_i b_i^\top,$$

where  $N \in \mathbb{R}^{p \times q}$  is the diffuse component and  $S$  is deterministic with fixed rank  $r = \text{rank}(S)$  as  $p, q \rightarrow \infty$ . The vectors  $a_i \in \mathbb{R}^p$  and  $b_i \in \mathbb{R}^q$  are orthonormal families and  $\beta_1 \geq \dots \geq \beta_r > 0$ . The dimensionless additive spike strengths are  $\theta_i = \beta_i^2 / (p\sigma^2)$ , and we consider sequences on which these strengths have fixed positive limits. Here  $\sigma^2$  is the per-entry variance of  $N$  from Assumption 1.

**Assumption 1** (Ideal Gaussian diffuse bulk). The diffuse component has iid entries  $N_{jk} \sim \mathcal{N}(0, \sigma^2)$ .

Assumption 1 supplies the closed-form thresholds. We apply it to the increment because trained raw-weight spectra are often heavy-tailed or otherwise depart from a pure Marchenko–Pastur law [12]. Gaussianity and isotropy of the increment remain testable assumptions. Theorems 3.1, 3.2, and 3.5 apply when those assumptions hold; Remark 3.6 gives the qualitative non-white alternative.

**Assumption 2** (Structure hypothesis, to be tested). A misalignment-relevant fine-tune deposits its signal in low rank,  $r_m$  small, while a benign fine-tune of equal energy spreads a comparable Frobenius budget over higher rank,  $r_b \gg r_m$ .

We treat Assumption 2 as an empirical hypothesis and condition the separation results below on it.

**Remark 4.2** (Fixed-energy comparison). By Definition 4.1, the signal contributes  $\|S\|_F^2 = \sum_i \beta_i^2 = p\sigma^2 \sum_i \theta_i$  to the energy. Holding the signal energy fixed therefore fixes  $\tau := \sum_i \theta_i$ . The discriminator of Section 5 concerns how that  $\tau$  is distributed across  $r$  directions. It assumes equal latent signal energy. Matching the observed  $\|\Delta W\|_F$  values in Section 2 is only a proxy unless the diffuse components are also controlled.

## 5 Main results

### 5.1 Detectability

**Proposition 5.1** (Detectability and location of a single spike). *Under Assumption 1 and the decomposition of Definition 4.1 with a single signal direction of induced strength  $\theta$ , the leading eigenvalue of  $C = \frac{1}{p}\Delta W^\top \Delta W$  satisfies, as  $p, q \rightarrow \infty$  with  $q/p \rightarrow \gamma$ ,*

$$\lambda_1(C) \xrightarrow{a.s.} \begin{cases} \sigma^2(1+\theta)\left(1+\frac{\gamma}{\theta}\right), & \theta > \sqrt{\gamma}, \\ \sigma^2(1+\sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases}$$

Consequently the signal is asymptotically visible in the spectrum, that is  $\lambda_1(C)$  exceeds the bulk edge by a deterministic gap, if and only if  $\theta > \sqrt{\gamma}$ .

*Proof.* Apply Theorem 3.2 with  $\theta = \beta^2/(p\sigma^2)$  directly in the additive model.  $\square$

### 5.2 The discriminator: rank at fixed energy

For deterministic fixed  $r$ , the rectangular finite-rank theorem applies to each planted singular direction. The detection of *any* outlier is governed by  $\max_i \theta_i$ , because the leading singular value tracks the strongest direction.

**Proposition 5.2** (Rank-at-fixed-energy discriminator). *For a fixed rank  $r$  in the proportional limit, fix the signal energy through  $\tau = \sum_{i=1}^r \theta_i$ . Among rank- $r$  signals of this energy, the leading induced strength satisfies  $\max_i \theta_i \geq \tau/r$ , with equality when the energy is spread equally,  $\theta_i = \tau/r$ . Therefore:*

1. *An equal-energy, equal-strength rank- $r$  update is spectrally detectable (Proposition 5.1) if and only if*

$$\frac{\tau}{r} > \sqrt{\gamma} \iff r < r_\star, \quad r_\star := \frac{\tau}{\sqrt{\gamma}}.$$

2. *Consequently, a concentrated update with  $r_m < r_\star$  produces a supercritical spike above the bulk edge, while an energy-matched diffuse update that spreads its budget approximately uniformly over rank  $r_b \geq r_\star$  keeps every direction subcritical and is asymptotically invisible. The test separates a concentrated update from a diffuse one of equal energy; a high-rank update that nonetheless places a supercritical share of its energy in a single direction is still detected.*

*Proof.* The inequality  $\max_i \theta_i \geq \tau/r$  is the pigeonhole bound on a sum of  $r$  nonnegative terms, with equality at the uniform allocation  $\theta_i = \tau/r$ . For the uniform allocation, every spike has strength  $\tau/r$ , so by Proposition 5.1 a detached eigenvalue exists if and only if  $\tau/r > \sqrt{\gamma}$ , equivalently  $r < \tau/\sqrt{\gamma} = r_\star$ . For statement (2): when  $r_m < r_\star$  the leading strength exceeds  $\sqrt{\gamma}$  and a spike appears, while a uniform allocation over  $r_b \geq r_\star$  puts every strength at or below  $\sqrt{\gamma}$  and no eigenvalue detaches. The pigeonhole bound is one-sided, so invisibility requires the diffuse, near-uniform allocation rather than  $r_b \geq r_\star$  alone; a high-rank update concentrated in one direction has  $\max_i \theta_i > \sqrt{\gamma}$  and is detected.  $\square$

At fixed latent signal energy, rank supplies a one-sided detectability bound through  $r_\star = \tau/\sqrt{\gamma}$ ; exact separation additionally requires every spike in the diffuse update to remain subcritical. Under Assumption 2, the spectral test separates matched updates only when those conditions hold. Section 7 describes observable proxies for this conditional prediction.

*Remark 5.3* (Uniform allocation minimizes detectability). Spreading energy uniformly minimizes the leading strength and is therefore the hardest case to detect. If the uniform allocation is detectable, every allocation of the same energy is detectable. Exact uniformity is unnecessary provided every benign spike remains subcritical.

### 5.3 Critical rank for representative layers

The critical rank  $r_\star = \tau/\sqrt{\gamma}$  depends on the layer through its aspect ratio  $\gamma$  and on the fine-tune through the signal energy  $\tau$ . Table 1 reports  $\gamma$  and  $r_\star$  for the released Llama-3-8B architecture (hidden width 4096, feed-forward width 14336, 32 query heads, and eight key/value heads) [16], at two illustrative signal energies.

| Layer                      | shape ( $p \times q$ ) | $\gamma = q/p$ | $r_\star$ at $\tau = 4, 16$ |
|----------------------------|------------------------|----------------|-----------------------------|
| Attention $q/o$ projection | $4096 \times 4096$     | 1.00           | 4.0, 16.0                   |
| Attention $k/v$ projection | $4096 \times 1024$     | 0.25           | 8.0, 32.0                   |
| MLP up / gate projection   | $14336 \times 4096$    | 0.286          | 7.5, 29.9                   |
| MLP down projection        | $14336 \times 4096$    | 0.286          | 7.5, 29.9                   |

Table 1: Critical rank  $r_\star = \tau/\sqrt{\gamma}$  across layer shapes. Shapes are written as (rows  $\times$  columns) after the transpose of Section 2 that places  $q \leq p$ , so  $\gamma = q/p$  is the smaller dimension over the larger. An equal-strength rank- $r$  signal is detectable in a layer exactly when  $r < r_\star$  for that layer.

The rectangular key/value and feed-forward maps have smaller  $\gamma$  and therefore higher critical ranks than the square query/output maps. A rank-one adapter can induce emergent misalignment [18], demonstrating low algebraic rank, but its spectral detectability cannot be inferred without its normalized strength and an applicable diffuse-bulk model. Accordingly, the estimator in Section 6 scans layers and reports each standardized leading eigenvalue; under the model, detection is easiest where  $\gamma$  is small and signal energy is large.

### 5.4 Label-free identification

Under the ideal model, a detected spike also yields an estimate of its planted direction.

**Proposition 5.4** (Label-free direction estimate). *Under the conditions of Proposition 5.1 with  $\theta > \sqrt{\gamma}$ , the leading eigenvector  $\hat{v}_1$  of  $C$  satisfies*

$$|\langle \hat{v}_1, b \rangle|^2 \xrightarrow{a.s.} \frac{1 - \gamma/\theta^2}{1 + \gamma/\theta} > 0,$$

where  $b$  is the planted signal direction. Thus  $\hat{v}_1$  is asymptotically nontrivially correlated with the perturbation direction, but is not a consistent direction estimator at fixed  $\theta$  and  $\gamma$ .

*Proof.* Immediate from the right-singular-vector statement of Theorem 3.2.  $\square$

Steering or ablating  $\hat{v}_1$  tests its causal effect on behavior. This compares the spectral estimate with the convergent activation direction reported for emergent misalignment [19]; the candidate itself is obtained without alignment labels. Persona-vector results provide related evidence for low-dimensional trait control [21], but do not identify an emergent-misalignment direction.

## 5.5 A calibrated test

**Proposition 5.5** (Known-scale level- $\alpha$  edge test). *Let  $H_0$  be the pure-noise null,  $\Delta W = N$  with no signal (Assumption 1 with  $\theta = 0$ ), and suppose  $\sigma^2$  is known. Using the centering and scaling of Theorem 3.5, the test that rejects  $H_0$  when*

$$\frac{\lambda_1(C) - \mu_{p,q}}{s_{p,q}} > t_{1-\alpha}, \quad t_{1-\alpha} = \text{TW}_1 \text{ quantile},$$

*has asymptotic false-positive rate  $\alpha$ . Null eigenvalue fluctuations have scale  $p^{-2/3}$ , while the critical window in spike strength around  $\theta = \sqrt{\gamma}$  has scale  $p^{-1/3}$ .*

*Proof.* Under  $H_0$  the leading eigenvalue obeys the Tracy–Widom limit of Theorem 3.5, so the standardized statistic exceeds  $t_{1-\alpha}$  with probability  $\alpha$  in the limit. For  $\theta = \sqrt{\gamma} + \epsilon$ , expansion of the outlier location in Proposition 5.1 gives

$$\sigma^2(1 + \theta)(1 + \gamma/\theta) - \lambda_+ = \frac{\sigma^2}{\sqrt{\gamma}}\epsilon^2 + O(\epsilon^3).$$

Matching this deterministic gap to the  $p^{-2/3}$  edge scale gives  $\epsilon \asymp p^{-1/3}$ . Calibrated power inside that critical window requires a deformed-spike analysis beyond the pure-null Tracy–Widom approximation.  $\square$

Under the known-scale Gaussian null, Proposition 5.5 gives asymptotic type-I error control and the critical-window scale. Empirical use still requires matched benign calibration and behavioral validation.

## 5.6 A subspace observable for rotation

Scalar spectral statistics depend only on the eigenvalues. Post-training can also rotate representation geometry [28], motivating a complementary eigenspace observable.

**Definition 5.6** (Leading-subspace principal-angle distance [3]). For two models with covariances  $C, C'$ , let  $V_k, V'_k$  be the matrices of their top- $k$  eigenvectors and let  $\rho_1 \geq \dots \geq \rho_k$  be the singular values of  $V_k^\top V'_k$ , equal to the cosines of the principal angles between the two subspaces. Define the Grassmann distance

$$d_k(C, C') = \left( \sum_{j=1}^k \arccos^2(\rho_j) \right)^{1/2}.$$

*Remark 5.7* (Calibrating the rotation observable). The distance  $d_k$  lacks Proposition 5.5’s scalar null. Under two independent isotropic Gaussian null draws, the eigenvector bases are Haar-distributed [2] and independent leading directions are nearly orthogonal in high dimension. Matched fine-tunes are correlated, so the independent-Haar null does not apply. We compare the candidate with its energy-matched benign control and use the design-specific resampling of Section 6.2 only as a sensitivity analysis. The resulting tail fraction is descriptive unless the transformations are exchangeable under a stated null. An asymptotic law for  $d_k$  would require a joint model of matched increments.

The subspace observable compares matched eigenspaces without behavioral labels and measures reorientation of existing spectral structure.



## 6 A label-free estimator

### 6.1 The procedure

1. **Form the increment.** For each layer of interest compute  $\Delta W = W_{\text{ft}} - W_{\text{base}}$  and the covariance  $C = \frac{1}{p} \Delta W^\top \Delta W$ . Record the aspect ratio  $\gamma = q/p$ .
2. **Estimate the bulk.** Fit the noise level  $\sigma^2$  from the bulk of the spectrum (for example by matching the median eigenvalue to the Marchenko–Pastur median), rather than from the full trace, which a genuine outlier can inflate. Because Proposition 5.5 assumes a known scale, this plug-in estimate requires separate calibration.
3. **Screen for an edge excursion.** Report the plug-in standardized leading eigenvalue as a heuristic score, the ideal-model implied strength  $\hat{\theta}$  from Proposition 5.1, and the descriptive number of MP-edge exceedances. The exceedance count need not equal algebraic signal rank. A calibrated decision requires a known scale or separate validation of the plug-in null.
4. **Match energy.** When comparing a misaligned candidate to a benign control, rescale so  $\|\Delta W_{\text{m}}\|_F = \|\Delta W_{\text{b}}\|_F$  before comparing spectra, using total Frobenius energy as an observable magnitude control. This does not by itself satisfy Proposition 5.2’s equal-latent-signal-energy assumption.
5. **Recover and test the direction.** Use the leading empirical singular vector as a candidate residual-stream direction and test behavioral coupling by steering or ablation. An effect establishes sensitivity to that intervention, not the planted-spike decomposition or a unique mechanism.
6. **Check rotation.** Compute the subspace distance of Definition 5.6 between matched models as a descriptive measure of reorientation rather than a detector of latent signal.

### 6.2 Finite-size surrogate stress tests

Proposition 5.5 provides asymptotic calibration for a Gaussian null with known scale. Real increments are finite, non-Gaussian, correlated, and analyzed with a fitted scale, so the plug-in Tracy–Widom statistic is a visibility heuristic. As a sensitivity check, independently permuting all entries of  $\Delta W$  preserves the global empirical entry multiset while destroying row and column structure. Independent sign flips preserve magnitudes and require a justified symmetric-sign null to preserve the signed marginal. For  $B$  surrogates, report the corrected tail fraction

$$\frac{1 + \#\{b : \lambda_1(C^{(b)}) \geq \lambda_1(C^{\text{obs}})\}}{B + 1}.$$

Under a stated null that makes the transformations exchangeable, this quantity is a randomization  $p$ -value; otherwise it is a descriptive surrogate tail fraction. Both transformations discard optimizer, architectural, row, and column correlations. The Llama-3-8B census therefore measures MP-visible structure but does not establish significance under an empirical null.

### 6.3 The activation-covariance instrument

The same construction can be applied formally to the activation covariance  $\Sigma_{\text{act}} = \frac{1}{n} \sum_t h_t h_t^\top$  of residual-stream states  $h_t$  over  $n$  probe tokens for matched models. In one reported rank-one-LoRA experiment, the layer-24 adapter  $B$  vector and a mean-difference activation direction have cosine

near 0.04, yet ablating that direction substantially suppresses the measured behavior [19]. This result motivates using activation covariance for identification.

For detection, activation covariance falls outside Propositions 5.1–5.5 because it is strongly anisotropic before fine-tuning [37, 38, 39]. Assumption 1 and the resulting closed forms therefore do not apply. A generalized spiked population model [10] offers an asymptotic analogue, but correlated token states need not satisfy its independent-sample premise. Detection would then depend on a less stable estimated deformed edge. Testing isotropy for weight increments does not validate it for activation covariance.

## 6.4 Weight-update rank and activation effective rank

A recent result compares models fine-tuned on the top- versus bottom-ranked subsets of nominally benign datasets and reports higher effective rank for the top-ranked, safety-degrading subset when processing harmful prompts [27]. That result concerns a different rank observable from the weight-increment rank studied here.

Effective rank [15] is  $\exp(H)$  where  $H = -\sum_i \pi_i \log \pi_i$  is the entropy of the normalized spectrum  $\pi_i = \lambda_i / \sum_j \lambda_j$ . Adding a single dominant spike to a covariance concentrates the distribution  $\pi$ , lowers  $H$ , and therefore *lowers* effective rank. Thus a low-rank misalignment signal lowers effective rank in the relevant covariance.

The cited study measures inference-time activation effective rank after fine-tuning on harmful *content*, without energy control; our prediction concerns the weight increment  $\Delta W$ . A concentrated update can still broaden the range of behaviors expressed on triggering inputs, so the two observables need not share a sign. Both should therefore be reported.

## 6.5 Spike confounds

Several artifacts can produce a leading-eigenvalue excursion unrelated to misalignment.

- **Outlier coordinates.** A few residual-stream coordinates with very large variance [37] create a top eigenvalue unrelated to any fine-tune. Control by per-coordinate standardization or robust covariance before spectral analysis, and by working on  $\Delta W$  where the effect is weaker.
- **Energy leakage.** Without the energy matching of step 4, the statistic may reflect training magnitude rather than misalignment.
- **Aspect-ratio regime.** For very wide matrices or many probe tokens,  $\gamma \rightarrow 0$ , the bulk collapses to a point and *every* spike is nominally supercritical ( $\sqrt{\gamma} \rightarrow 0$ ). Detectability is then trivial, and benign-versus-misaligned separation depends entirely on the rank argument of Proposition 5.2. Report  $\gamma$  explicitly and interpret the threshold only where it is nontrivial.
- **Heavy-tailed bulk.** Many trained raw-weight spectra depart from a pure Marchenko–Pastur law [12]. The Gaussian model is therefore posed on  $\Delta W$ , but its bulk fit must still be checked. If it deviates, discard the MP closed form and use Remark 3.6 only as a qualitative guide.

## 7 Empirical test

**Matched-concentration test.** Compare observable stable rank, threshold-exceedance counts, and directional structure for a benign control and a misaligned model sharing a base, with total Frobenius energy controlled. These statistics test concentration but do not identify  $r = \text{rank}(S)$  or  $\tau = \|S\|_F^2 / (p\sigma^2)$ . Locating a checkpoint relative to  $r_\star = \tau / \sqrt{\gamma}$  would require an explicit, validated

estimator for the latent decomposition. If the observable summaries do not separate the matched models, they provide no scalar support for Assumption 2.

**Substrate.** The matched comparison requires *full* fine-tuning. A rank-constrained adapter fixes the rank of  $\Delta W$  by construction, precluding a rank difference, and its increment has no bulk for a spike to cross. Emergent-misalignment organisms exist as full fine-tunes and as rank-one adapters [17, 18]; the adapters provide controlled intervention examples but cannot test Propositions 5.1–5.4 without a diffuse bulk. The spectral claims of Propositions 5.1–5.2 require full fine-tunes with a matched benign control trained under an identical recipe.

**Prediction.** Under the rank-structure antecedent of Assumption 2, the ideal scalar prediction is a supercritical leading eigenvalue in  $\Delta W$  at a layer where the benign control remains subcritical. If benign fine-tuning is also anisotropic, evidence must instead come from contrastive singular-vector comparison, behavioral ablation, and calibrated same-recipe held-out screens. If the measurements do not separate the models, the corresponding part of Assumption 2 has failed.

## 8 Limitations

1. **Rank separation at matched energy.** If misaligned and benign fine-tunes are equally concentrated at matched energy, rank does not discriminate and the method has no signal. Assumption 2 makes this condition directly measurable.
2. **Activation anisotropy.** Activation covariance violates the isotropic-bulk model used for  $\Delta W$  and would require a deformed-bulk analysis beyond the scope of this paper (Section 6.3).
3. **Heavy tails on raw weights.** We place the Marchenko–Pastur null on the diffuse part of the increment. Departures from MP in trained raw-weight spectra [12] motivate avoiding a direct threshold on  $W$  but do not establish the null for increments.
4. **Narrow versus broad misalignment.** Narrow fine-tuning can induce broad behavioral misalignment [17]; whether broader behavioral effects correspond to more concentrated weight updates remains unknown.
5. **Internalized and deceptive objectives.** Belief–statement probes catch instrumental deception but largely miss internalized objectives, which present no such gap [36]. We target the internalized case, where behavioral probes are weakest and no belief-gap signal is available. Any spectral signal would be structural and present in the weights regardless of whether the model ever “intends” to deceive.
6. **Adaptive evasion.** An adaptive model could spread the update over higher rank or move it into a benign-looking subspace, as demonstrated for other latent-space monitors [35]. Robustness to such evasion remains untested.
7. **Finite-size and regime gaps.** The limits are asymptotic, whereas real layers have finite  $p, q$ , and treating the rows of a single increment as observations requires justification. Tracy–Widom calibration further assumes Gaussianity and known scale. Without finite-matrix calibration, the fitted-scale screen remains heuristic.
8. **Layerwise versus model-level validity.** The propositions concern one prespecified matrix. Scanning layers or map types requires calibration for multiple dependent tests; an excursion in one matrix is not a calibrated model-level detector.

9. **Training-process and scale scope.** The setup models one endpoint increment from a fixed base. It does not separate optimizer effects from SFT, RLHF, or DPO; the  $p, q \rightarrow \infty$  limit concerns matrix dimensions, not transfer across model sizes or architectures. Sequential or continual fine-tuning requires new reference increments and matched calibration.

## 9 Related work

**Spectral fingerprints of models.** Heavy-tailed self-regularization reads model quality from weight spectra without data or labels, and names a “bulk plus spikes” phase [12]. We apply this vocabulary to misalignment in the weight increment, using the rectangular additive threshold rather than a power-law tail index. Work connecting weight spectra, activation covariance, and transformer fine-tuning finds refinement along small-singular-value directions [14]; our matched-energy comparison asks whether benign and misaligned increments differ. Spectral fingerprints also support model lineage detection designed to be invariant to fine-tuning [40]. Random-matrix descriptions of network spectra appear in analyses of loss-surface Hessians [11] and local spectral statistics of trained weight matrices, which remain random-matrix-like even when the global density does not [13].

**Spectral and random-matrix detection.** Spectral outlier scores for data poisoning [30] and random-matrix hallucination detectors [29] operate per input. The present statistic operates per model and uses rank at fixed energy.

**The structure of emergent misalignment.** Emergent misalignment, in which narrow fine-tuning induces broad misalignment [17], is mediated by a single convergent linear direction in activations [19], can be induced in clean model organisms by a rank-one adapter [18], is affected by persona features [20, 21], and has reported shared low-dimensional parameter subspaces across tasks [22]. Together, these results motivate Assumption 2 without establishing it. Under that assumption, the rectangular additive model predicts an outlier transition controlled by rank at fixed energy. The single-direction phenomenon is part of a broader pattern in which safety-relevant behaviors can be low-dimensional, as for refusal [23]; fine-tuning objectives can be optimized in low-intrinsic-dimensional parameter subspaces [24], and explicitly low-rank adapters are effective [25].

**Geometry of fine-tuning and safety.** A complementary line analyzes how fine-tuning breaks safety through low-rank curvature subspaces and argues that benign tasks can also degrade alignment [26], making the benign-side measurement essential. Effective-rank and spectral-slope measurements likewise show that post-training changes representation geometry [28]. Our analysis concerns detection between matched models.

**Behavioural and label-based detection.** Supervised probes detect deception at high accuracy when labels and the distribution are available [31], and interpretability-driven audits surface hidden objectives given sufficient access [32]. Unsupervised methods that search for latent knowledge exist [33], but arbitrary features can satisfy their consistency constraints [34]. We focus on label-free detection under distributional uncertainty; when suitable labels and a known distribution are available, supervised probes should perform better.

## 10 Conclusion

In the fixed-rank additive model, equal-strength spikes at controlled latent signal energy cross the threshold exactly when  $r < r_* = \tau/\sqrt{\gamma}$ ; for arbitrary allocations, low rank gives only a one-sided guarantee. If Assumption 2 holds and every matched benign spike remains subcritical, the spectral statistic separates the models and estimates a candidate planted direction without behavioral labels. Applying this result to real fine-tunes requires validated estimates of the latent decomposition; stable rank and edge-exceedance counts are concentration proxies, not estimates of  $r$  or  $\tau$ .

## References

- [1] V. A. Marčenko and L. A. Pastur, “Distribution of eigenvalues for some sets of random matrices,” *Math. USSR-Sbornik* **1**(4):457–483, 1967.
- [2] Z. Bai and J. W. Silverstein, *Spectral Analysis of Large Dimensional Random Matrices*, 2nd ed., Springer, 2010.
- [3] A. Edelman, T. A. Arias, and S. T. Smith, “The geometry of algorithms with orthogonality constraints,” *SIAM J. Matrix Anal. Appl.* **20**(2):303–353, 1998. doi:10.1137/S0895479895290954.
- [4] I. M. Johnstone, “On the distribution of the largest eigenvalue in principal components analysis,” *Ann. Statist.* **29**(2):295–327, 2001.
- [5] J. Baik, G. Ben Arous, and S. Péché, “Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices,” *Ann. Probab.* **33**(5):1643–1697, 2005.
- [6] J. Baik and J. W. Silverstein, “Eigenvalues of large sample covariance matrices of spiked population models,” *J. Multivariate Anal.* **97**(6):1382–1408, 2006.
- [7] D. Paul, “Asymptotics of sample eigenstructure for a large dimensional spiked covariance model,” *Statist. Sinica* **17**:1617–1642, 2007.
- [8] F. Benaych-Georges and R. R. Nadakuditi, “The singular values and vectors of low rank perturbations of large rectangular random matrices,” *J. Multivariate Anal.* **111**:120–135, 2012.
- [9] C. A. Tracy and H. Widom, “On orthogonal and symplectic matrix ensembles,” *Comm. Math. Phys.* **177**(3):727–754, 1996.
- [10] Z. Bai and J. Yao, “On sample eigenvalues in a generalized spiked population model,” *J. Multivariate Anal.* **106**:167–177, 2012.
- [11] J. Pennington and Y. Bahri, “Geometry of neural network loss surfaces via random matrix theory,” *ICML*, PMLR 70:2798–2806, 2017.
- [12] C. H. Martin and M. W. Mahoney, “Implicit self-regularization in deep neural networks: evidence from random matrix theory and implications for learning,” *J. Mach. Learn. Res.* **22**(165):1–73, 2021.
- [13] M. Thamm, M. Staats, and B. Rosenow, “Random matrix analysis of deep neural network weight matrices,” *Phys. Rev. E* **106**:054124, 2022.
- [14] M. Staats, M. Thamm, and B. Rosenow, “Small singular values matter: a random matrix analysis of transformer models,” *NeurIPS*, vol. 38, 2025. arXiv:2410.17770.

- [15] O. Roy and M. Vetterli, “The effective rank: a measure of effective dimensionality,” *European Signal Processing Conference (EUSIPCO)*, 606–610, 2007.
- [16] A. Grattafiori et al., “The Llama 3 herd of models,” arXiv preprint arXiv:2407.21783, 2024.
- [17] J. Betley, D. C. H. Tan, N. Warncke, A. Sztyber-Betley, X. Bao, M. Soto, N. Labenz, and O. Evans, “Emergent misalignment: narrow finetuning can produce broadly misaligned LLMs,” *ICML*, PMLR 267:4043–4068, 2025. arXiv:2502.17424.
- [18] E. Turner, A. Soligo, M. Taylor, S. Rajamanoharan, and N. Nanda, “Model organisms for emergent misalignment,” arXiv:2506.11613, 2025.
- [19] A. Soligo, E. Turner, S. Rajamanoharan, and N. Nanda, “Convergent linear representations of emergent misalignment,” arXiv:2506.11618, 2025.
- [20] M. Wang, T. Dupré la Tour, O. Watkins, A. Makelov, R. A. Chi, S. Miserendino, J. Wang, A. Rajaram, J. Heidecke, T. Patwardhan, and D. Mossing, “Persona features control emergent misalignment,” *ICLR*, 2026. arXiv:2506.19823.
- [21] R. Chen, A. Ardit, H. Sleight, O. Evans, and J. Lindsey, “Persona vectors: monitoring and controlling character traits in language models,” arXiv:2507.21509, 2025.
- [22] D. A. R. Arturi, E. Zhang, A. Ansah, K. Zhu, A. Panda, and A. Balwani, “Shared parameter subspaces and cross-task linearity in emergently misaligned behavior,” arXiv:2511.02022, 2025.
- [23] A. Ardit, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda, “Refusal in language models is mediated by a single direction,” *NeurIPS*, 2024. arXiv:2406.11717.
- [24] A. Aghajanyan, S. Gupta, and L. Zettlemoyer, “Intrinsic dimensionality explains the effectiveness of language model fine-tuning,” *ACL-IJCNLP*, 2021. arXiv:2012.13255.
- [25] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: low-rank adaptation of large language models,” *ICLR*, 2022. arXiv:2106.09685.
- [26] M. Springer, C. P. Lee, B. Metevier, J. Castleman, B. Turbal, H. Jung, Z. Shen, and A. Korolova, “The geometry of alignment collapse: when fine-tuning breaks safety,” arXiv:2602.15799, 2026. Preprint.
- [27] H. Li, L. Li, Z. Lu, X. Wei, R. Li, J. Shao, and L. Sha, “Layer-aware representation filtering: purifying finetuning data to preserve LLM safety alignment,” *EMNLP*, 8030–8050, 2025. doi:10.18653/v1/2025.emnlp-main.406.
- [28] M. Z. Li, K. K. Agrawal, A. Ghosh, K. K. Teru, A. Santoro, G. Lajoie, and B. A. Richards, “Tracing the representation geometry of language models from pretraining to post-training,” *NeurIPS*, vol. 38, 2025. arXiv:2509.23024.
- [29] D. Ettore, “Spectral Geometry for Deep Learning: Compression and Hallucination Detection via Random Matrix Theory,” arXiv:2601.17357, 2026. MS thesis, University of Illinois Chicago, defended November 21, 2025.
- [30] B. Tran, J. Li, and A. Małdry, “Spectral signatures in backdoor attacks,” *NeurIPS*, 2018. arXiv:1811.00636.

- [31] N. Goldowsky-Dill, B. Chughtai, S. Heimersheim, and M. Hobbhahn, “Detecting strategic deception using linear probes,” arXiv:2502.03407, 2025.
- [32] S. Marks, J. Treutlein, T. Bricken, J. Lindsey, J. Marcus, S. Mishra-Sharma, D. Ziegler, E. Ameisen, J. Batson, T. Belonax, S. R. Bowman, S. Carter, B. Chen, H. Cunningham, C. Denison, F. Dietz, S. Golechha, A. Khan, J. Kirchner, J. Leike, A. Meek, K. Nishimura-Gasparian, E. Ong, C. Olah, A. Pearce, F. Roger, J. Salle, A. Shih, M. Tong, D. Thomas, K. Rivoire, A. Jermyn, M. MacDiarmid, T. Henighan, and E. Hubinger, “Auditing language models for hidden objectives,” arXiv:2503.10965, 2025.
- [33] C. Burns, H. Ye, D. Klein, and J. Steinhardt, “Discovering latent knowledge in language models without supervision,” *ICLR*, 2023. arXiv:2212.03827.
- [34] S. Farquhar, V. Varma, Z. Kenton, J. Gasteiger, V. Mikulik, and R. Shah, “Challenges with unsupervised LLM knowledge discovery,” arXiv:2312.10029, 2023.
- [35] L. Bailey, A. Serrano, A. Sheshadri, M. Seleznyov, J. Taylor, E. Jenner, J. Hilton, S. Casper, C. Guestrin, and S. Emmons, “Obfuscated activations bypass LLM latent-space defenses,” *ICLR*, 2026. arXiv:2412.09565.
- [36] K. Haralambiev, “Why safety probes catch liars but miss fanatics,” arXiv:2603.25861, 2026. Preprint.
- [37] M. Sun, X. Chen, J. Z. Kolter, and Z. Liu, “Massive activations in large language models,” *COLM*, 2024. arXiv:2402.17762.
- [38] J. Gao, D. He, X. Tan, T. Qin, L. Wang, and T.-Y. Liu, “Representation degeneration problem in training natural language generation models,” *ICLR*, 2019. arXiv:1907.12009.
- [39] K. Ethayarajh, “How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings,” *EMNLP-IJCNLP*, 2019.
- [40] S. Wang, Z. Ma, X. Li, and Z. Li, “Ghost in the Transformer: Detecting Model Reuse with Invariant Spectral Signatures,” *Proc. AAAI Conf. Artif. Intell.* **40**(40):33648–33656, 2026. doi:10.1609/aaai.v40i40.40654. arXiv:2511.06390.